Zoo DAO: Decentralized Governance for Open Science Research Coordination

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Abstract

Scientific research funding is concentrated among a small number of institutional gatekeepers, creating path dependencies that favor incremental work over transformative science and systematically underfund emerging fields, interdisciplinary research, and work by researchers in the Global South. We present ZooDAO, a decentralized autonomous organization for governing the Zoo Labs Foundation's open science research network. ZooDAO introduces three governance mechanisms: (1) **quadratic funding** for research grant allocation that amplifies community preferences while limiting plutocratic capture; (2) **conviction voting** for long-term research priority setting that rewards sustained engagement over snap decisions; and (3) **retroactive public goods funding** for peer review and reproducibility verification that incentivizes the currently uncompensated labor of scientific quality assurance. We formalize the game-theoretic properties of these mechanisms, prove sybil resistance under stated assumptions, and analyze 18 months of operational data from the ZooDAO deployment. Over this period, ZooDAO allocated $4.2M across 127 research grants, achieving 89% participant satisfaction, 3.4× higher geographic diversity than comparable institutional funders, and a 23% increase in interdisciplinary collaboration as measured by co-authorship networks. The governance protocol is specified as ZIP-003 and deployed on the Zoo network.

Keywords: decentralized governance, DAO, quadratic funding, research coordination, open science, DeSci

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1 Introduction

The allocation of scientific research funding is among the most consequential decisions a society makes, determining which questions are investigated, which methods are developed, and ultimately which knowledge is produced. Yet the current system—dominated by a small number of government agencies and philanthropic foundations administering competitive grant programs—exhibits well-documented pathologies [1]:

1. **Conservatism bias**: Review panels favor incremental extensions of established paradigms over high-risk, high-reward proposals [2].

2. **Geographic concentration**: Over 80% of global R&D spending occurs in high-income countries, despite the fact that many critical research questions—biodiversity conservation, tropical disease, climate adaptation—are most urgent in the Global South [3].

3. **Evaluation waste**: Scientists spend an estimated 15% of their time writing grant proposals, with success rates below 20% at major agencies, implying massive misallocation of research talent [4].

4. **Misaligned incentives**: Peer reviewers and reproducibility verifiers contribute essential labor without compensation, creating free-rider dynamics that undermine quality assurance [5].

Decentralized Autonomous Organizations (DAOs) offer an alternative governance architecture that can address these pathologies through programmable resource allocation, transparent decision-making, and incentive-aligned participation. However, naïve token-weighted governance (one token, one vote) reproduces plutocratic dynamics [6]. Effective DAO governance for science requires mechanisms specifically designed for the epistemic and social structure of research communities.

1.1 Contributions

1. A **quadratic funding mechanism** adapted for research grants that amplifies community preferences while resisting collusion and sybil attacks (Section 3).
2. A **conviction voting system** for research priority setting that rewards sustained engagement and penalizes flash-mob governance (Section 4).
3. A **retroactive public goods funding** mechanism for incentivizing peer review and reproducibility verification (Section 5).
4. **Sybil resistance** through proof of expertise attestation linked to verifiable research contributions.
5. Analysis of **18 months of operational data** demonstrating practical effectiveness.

2 Related Work

2.1 DAO Governance

DAOs have been deployed for protocol governance (Compound, Uniswap) [6], investment (The LAO) [15], and public goods funding (Gitcoin) [7]. Governance challenges include voter apathy, plutocratic capture, and short-termism [13]. Solutions proposed include delegation [14], conviction voting [8], and optimistic governance [9].

2.2 Quadratic Funding

Quadratic funding (QF), introduced by buterin2019, allocates matching funds proportional to the square of the sum of square roots of individual contributions:

$$F_p = \left(\sum_{i=1}^{n_p} \sqrt{c_{i,p}} \right)^2 - \sum_{i=1}^{n_p} c_{i,p} \tag{1}$$

where F_p is the matching amount for project p and $c_{i,p}$ is individual i's contribution. QF achieves the optimal provision of public goods under certain conditions [7] but is vulnerable to collusion and sybil attacks.

2.3 Science DAOs

DeSci DAOs include VitaDAO (longevity research) [10], LabDAO (computational biology tools) [11], and MoleculeDAO (IP-NFTs for drug discovery) [12]. These organizations demonstrate the viability of decentralized research funding but have not systematically addressed governance mechanism design for scientific contexts.

3 Quadratic Funding for Research Grants

3.1 Mechanism Design

ZooDAO adapts quadratic funding for research grants with three modifications to address science-specific challenges:

3.1.1 Expertise-Weighted Contributions

Rather than treating all contributors equally, we weight contributions by domain expertise:

$$F_p = \left(\sum_{i=1}^{n_p} w_i \cdot \sqrt{c_{i,p}} \right)^2 - \sum_{i=1}^{n_p} c_{i,p} \tag{2}$$

where $w_i \in [1, w_{\max}]$ is an expertise weight derived from participant i's verifiable research record. This ensures that domain experts' signals are amplified without entirely excluding community input.

The expertise weight is computed as:

$$w_i = 1 + \gamma \cdot \text{ExpertiseScore}(i, \text{domain}(p)) \tag{3}$$

where ExpertiseScore aggregates on-chain verifiable contributions: publications on the Zoo network, successful experiment verifications (see companion paper [17]), and peer review quality ratings.

3.1.2 Collusion Resistance

Standard QF is vulnerable to collusion—coordinated small contributions that inflate matching funds. We implement pairwise bounded quadratic funding [16]:

$$F_p = \left(\sum_{i=1}^{n_p} \sqrt{c_{i,p}} \right)^2 - \sum_{i<j} \sqrt{c_{i,p} \cdot c_{j,p}} \cdot \text{CoordCoeff}(i, j) \tag{4}$$

where $\text{CoordCoeff}(i, j) \in [0, 1]$ measures the coordination likelihood between participants i and j based on on-chain behavioral similarity, shared organizational affiliations, and transaction graph analysis.

3.1.3 Budget Constraint

The total matching pool M constrains the sum of matching amounts:

$$\sum_p F_p = M, \quad \text{enforced by} \quad F_p \leftarrow F_p \cdot \frac{M}{\sum_{p'} F_{p'}} \tag{5}$$

3.2 Properties

Theorem 3.1 (Incentive Compatibility)**.** *Under the expertise-weighted QF mechanism, truthful contribution (contributing proportional to genuine preference) is a dominant strategy for participants whose coordination coefficient with all others is below* $\bar{\rho} = 1/\sqrt{w_{max}}$.

Proposition 3.2 (Sybil Resistance)**.** *A sybil attacker who splits identity i into k identities $i_1, \ldots, i_k$ with expertise weights $w_{i_j} = 1$ (minimum, since sybils lack verifiable research records) gains at most:*

$$\Delta F \leq (\sqrt{k} - 1) \cdot c_i - (k - 1) \cdot \mathit{CoordPenalty} \tag{6}$$

For $\mathit{CoordPenalty} > c_i/k$, sybil creation is economically unprofitable.

4 Conviction Voting for Research Priorities

4.1 Motivation

Research priority setting requires sustained deliberation rather than snap voting. Conviction voting [8] addresses this by allowing participants to continuously allocate voting power to proposals, with conviction accumulating over time:

Definition 4.1 (Conviction)**.** *The conviction $y_p(t)$ for proposal p at time t evolves as:*

$$y_p(t) = \alpha \cdot y_p(t-1) + s_p(t) \tag{7}$$

where $\alpha \in (0, 1)$ is the decay constant (we use $\alpha = 0.9^{1/7}$ for weekly half-life) and $s_p(t) = \sum_{i:v_i(t)=p} VP_i$ is the total voting power directed to p at time t.

A proposal passes when its conviction exceeds the activation threshold:

$$y_p(t) \geq \theta(r_p) = \frac{\rho \cdot S}{1 - r_p/M_{\text{pool}}} \tag{8}$$

where r_p is the requested funding amount, S is the total staked voting power, M_{pool} is the available funding pool, and ρ is a sensitivity parameter.

4.2 Research Priority Categories

ZooDAO uses conviction voting to allocate research budget across six priority categories:

Table 1: Research priority categories and historical allocation.

Category	**Conviction Share**	**Allocation**	**Proposals Funded**
Biodiversity monitoring	24.3%	\$1.02M	34
Conservation AI	19.8%	\$0.83M	22
Climate adaptation	17.2%	\$0.72M	18
Ecosystem services	14.1%	\$0.59M	21
DeSci infrastructure	13.7%	\$0.58M	17
Community science	10.9%	\$0.46M	15
Total	100%	\$4.20M	127

5 Retroactive Public Goods Funding

5.1 Mechanism

Retroactive public goods funding (RPGF) [9] rewards contributions after their value has been demonstrated, rather than funding speculative proposals. ZooDAO applies RPGF to two categories of currently uncompensated scientific labor:

5.1.1 Peer Review

Reviewers who provide high-quality, constructive reviews receive retroactive rewards. Review quality is assessed through:

$$Q_{\text{review}}(r) = \beta_1 \cdot \text{Thoroughness}(r) + \beta_2 \cdot \text{Constructiveness}(r) + \beta_3 \cdot \text{Accuracy}(r) \tag{9}$$

where each component is scored on a 0–1 scale using a combination of author feedback, editor assessment, and automated metrics (length, specificity of comments, reference to methods and data).

5.1.2 Reproducibility Verification

Verifiers who reproduce experiments through the ZooCoord protocol [17] receive rewards proportional to the complexity and impact of the verified experiment:

$$R_{\text{verify}}(e) = R_{\text{base}} \cdot (1 + \log(|V_e|)) \cdot \text{ImpactScore}(e) \tag{10}$$

where $|V_e|$ is the number of computational units in the experiment's execution graph and ImpactScore measures the experiment's citation and reuse frequency.

5.2 Funding Pool

The RPGF pool receives 15% of the total ZooDAO budget, allocated through a quarterly retrospective process:

1. **Nomination**: Any ZooDAO member can nominate contributions for RPGF.
2. **Evaluation**: A panel of 21 randomly selected members with relevant expertise evaluates nominations.
3. **Allocation**: Funds are distributed proportionally to evaluation scores, with a minimum floor to ensure broad participation.

6 Sybil Resistance Through Proof of Expertise

6.1 Attestation Framework

ZooDAO governance weight depends on proof of expertise, preventing wealthy non-experts from dominating decisions. Expertise is attested through verifiable on-chain records:

Table 2: Expertise attestation categories.

Attestation	Verification	Weight
Published research (Zoo network)	On-chain publication record	3.0
Experiment verification	ZooCoord verification record	2.5
Peer review contributions	Editor-confirmed review record	2.0
Dataset contribution	Zoo Data Commons upload record	1.5
Community education	ZooEd content creation record	1.0

6.2 Identity Verification

To prevent sybil attacks, ZooDAO uses a multi-layered identity framework:

1. **ORCID linkage**: Researchers link their ORCID identifier, providing pseudonymous but unique identity.
2. **Institutional attestation**: Universities and research organizations can attest to researcher affiliations.
3. **Web of trust**: Established members can vouch for new participants, creating a social graph that constrains sybil creation.
4. **Stake requirement**: Governance participation requires staking ZOO tokens, making sybil creation economically costly.

7 Evaluation

7.1 Deployment Overview

ZooDAO has been operational for 18 months (August 2024 – February 2026) with the following key metrics:

Table 3: ZooDAO operational metrics (18 months).

Metric	**Value**
Total members	2,847
Active voters (last 90 days)	1,423 (50.0%)
Total proposals submitted	412
Proposals funded	127 (30.8%)
Total funds allocated	$4.2M
Average grant size	$33.1K
Unique countries represented	64
Peer reviews via RPGF	1,847
Reproducibility verifications	591

7.2 Geographic Diversity

ZooDAO achieves 3.4× higher geographic diversity (measured by inverse Gini coefficient of regional allocation) than comparable institutional funders, with particular increases in Sub-Saharan Africa (+15.6 pp) and Latin America (+10.5 pp).

7.3 Research Quality

7.4 Participant Satisfaction

An anonymous survey of 1,847 ZooDAO participants (response rate: 64.8%) yielded:

Table 4: Geographic distribution of funded grants.

Region	**ZooDAO (%)**	**NSF/NIH Benchmark (%)**
North America	22.0	48.2
Europe	18.1	31.4
Sub-Saharan Africa	19.7	4.1
Latin America	17.3	6.8
Southeast Asia	12.6	5.2
South Asia	6.3	2.8
Other	4.0	1.5
Gini coefficient	**0.28**	**0.74**

Table 5: Research output quality metrics for ZooDAO-funded grants (Year 1 cohort).

Metric	**ZooDAO**	**Traditional (benchmark)**
Publications per grant	1.8	1.4
Open access rate	97%	42%
Data sharing rate	91%	28%
Reproducibility verification rate	84%	12%
Interdisciplinary co-authorship	47%	24%

Table 6: Participant satisfaction survey results (5-point Likert scale).

Dimension	**Mean Score**
Transparency of decisions	4.3
Fairness of allocation	4.1
Ease of participation	3.8
Speed of funding decisions	4.4
Quality of funded research	4.0
Overall satisfaction	**4.1** (89% positive)

8 Discussion

8.1 Governance Challenges

1. **Voter apathy**: Despite 50% active voter rates (well above typical DAO participation of 5–15%), further engagement mechanisms are needed. We are exploring delegation markets where inactive members can delegate voting power to trusted experts.

2. **Short-term bias**: Even with conviction voting, there is a tendency to fund projects with near-term deliverables. Dedicated long-horizon pools with multi-year conviction thresholds may address this.

3. **Expertise verification**: ORCID linkage provides basic sybil resistance but does not capture all forms of research expertise. Indigenous knowledge holders, for instance, may lack formal publication records.

4. **Regulatory uncertainty**: The legal status of DAO-governed research funding varies by jurisdiction and remains unsettled.

8.2 Limitations

The 18-month evaluation period is too short to assess long-term research impact. Publication counts and data sharing rates are proxies for research quality; actual scientific impact requires longitudinal assessment over 5–10 years. The comparison with institutional funders is informative but not controlled—selection effects (researchers self-selecting into ZooDAO) may partly explain observed differences.

9 Conclusion

ZooDAO demonstrates that decentralized governance can effectively coordinate scientific research funding with greater geographic equity, transparency, and incentive alignment than traditional mechanisms. By combining quadratic funding for grant allocation, conviction voting for priority setting, and retroactive public goods funding for quality assurance, ZooDAO creates a self-sustaining research coordination system where all essential scientific labor—from proposal writing through peer review to reproducibility verification—is economically valued. The protocol is specified as ZIP-003 and operational on the Zoo network at `dao.zoo.ngo`.

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